Formula 1 Podium Prediction

ACL Project, Milestone 1: analytical report

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This report collects the written results of our notebook in one place. Sections 2 and 3 hold the three parts the brief asks for: the answers to the data-engineering questions and the selected features with why each is available before the race. Section 8 lists the limitations we know about. Sections 1 and 4–7 summarise the cleaning, the models, the experiments, the explanations and the inference function, so the report can be read without the notebook. All numbers come from the notebook's outputs, and section numbers in brackets (e.g. [2.1]) point to the matching notebook section.

1. Data, cleaning and EDA

The source is the Formula 1 World Championship dataset (1950–2024), 14 relational CSV tables. The raw files are never modified; all cleaning happens on copies. We explored the data before cleaning, cleaned it, and then explored it again to see what the cleaning changed.

1.1 Before cleaning

The first finding was that the dataset hides its missing values. Every gap is written as the literal string `\N`, so pandas reported 0 missing values in raw `results` although there were 122,887. The same marker turned numeric columns such as position, milliseconds and rank into text. Several columns are almost entirely empty (practice-session dates in races above 90%, constructor_results.status 99.9%).

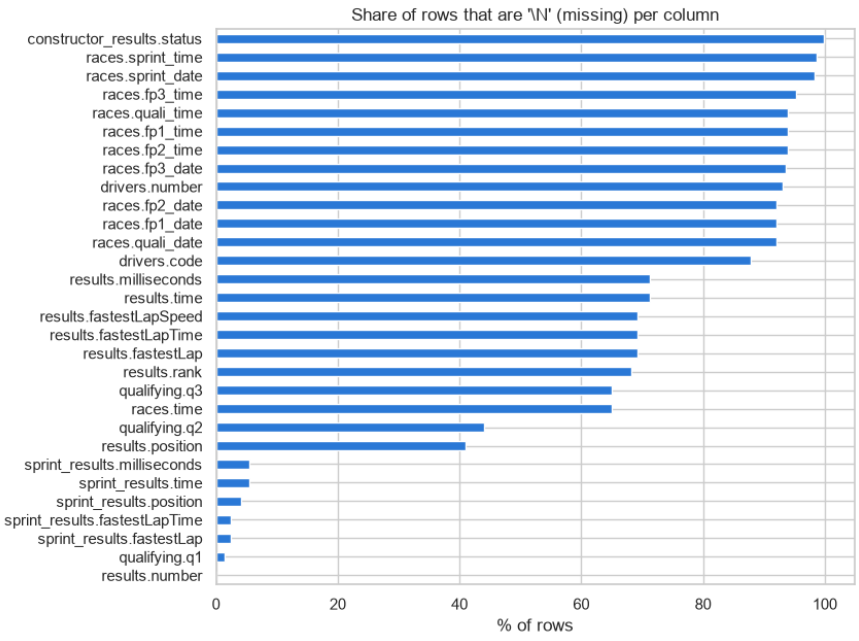


Figure 1. *Share of rows that are* `\N` *in each column, raw data. None of these are visible to pandas before cleaning.*

The target is imbalanced: only 12.7% of all entries are podiums. Starting position matters a lot. Pole gives a 63% podium rate, P3 gives 46%, and beyond P10 the rate is under 5%. The grid differs from the qualifying position for 23% of entries, mostly because of penalties, so the two are treated as separate things. 85 driver–race pairs (176 rows, all 1950–1978) appear more than once, and nobody ever reached the podium from grid = 0 (a pit-lane start or a non-qualifier).

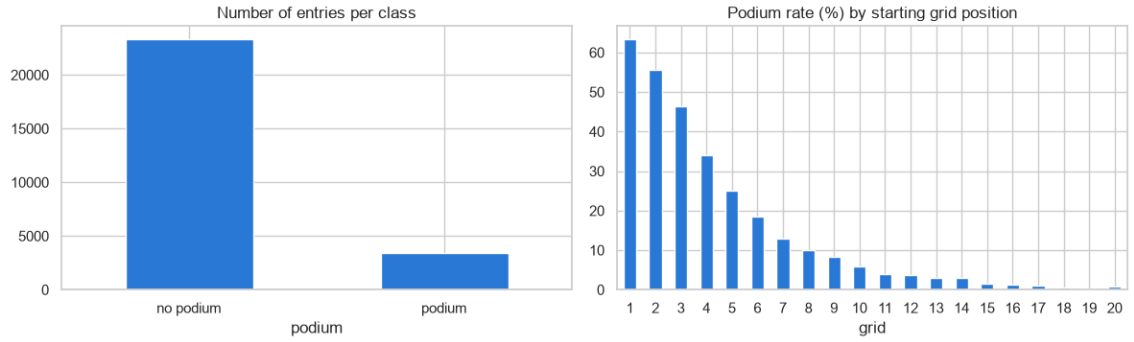


Figure 2. *Left: class balance over all entries. Right: podium rate by starting grid position.*

1.2 Cleaning

- Replaced 160,076 `\N` markers with real missing values, retyped 18 columns (numbers, dates, and lap and pit-stop times converted to seconds) and dropped 20 unused columns [1.2].
- Fixed 56 impossible or contradictory values and two category spellings, and set one impossible Q1 time (16:42.640 at the 1995 Japanese GP, a recording error) to missing [1.2.6–1.2.8].
- All 14 primary keys are unique and non-null, and every foreign key resolves to its parent table [1.3].
- For the 85 duplicated pairs, mostly shared drives where drivers swapped cars mid-race, we keep the row with the best `positionOrder`. This keeps all 22 podiums in those pairs (keeping the first row would lose 16), and the kept row is also the car the driver did most laps in for every pair. `results` now has exactly one row per (`raceId`, `driverId`): 26,759 to 26,668 rows [1.4].

1.3 After cleaning

Once the markers are replaced, the real gaps become visible: 103,336 in results and 11,669 in qualifying. Most are missing by design rather than errors. Q2 and Q3 times only exist for drivers who reached those sessions, race time only for drivers on the lead lap, and fastest-lap data was not recorded before the mid-2000s. Most of these columns are post-race information and are never used as features. The two columns we rely on most, `positionOrder` and `grid`, have no missing values.

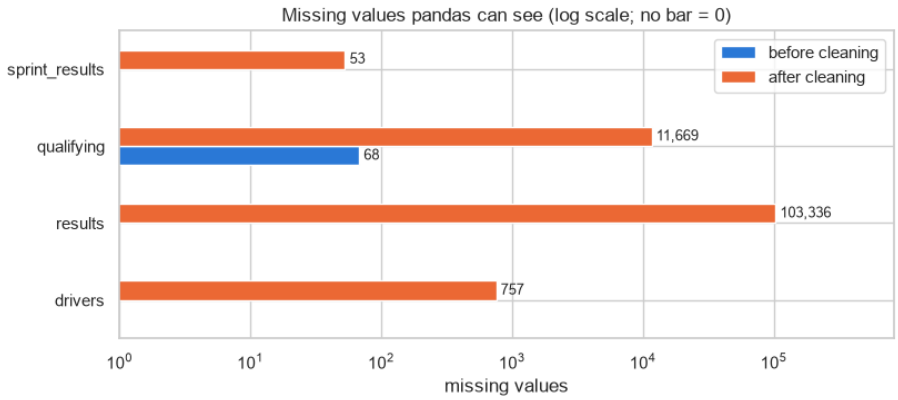


Figure 3. *Missing values pandas can see before and after cleaning (log scale; no bar means zero).*

Typed columns could finally be plotted. Raw Q1 times range from about 54 to 150 seconds because circuits differ in length, which is why the features use the gap to the fastest lap instead of raw times. Pit-stop durations have a cluster between 400 and 3,069 seconds; these are red-flag stoppages, which are real records and were kept. Driver age on race day runs from 17.5 to 58.8 years, which confirms the date parsing worked.

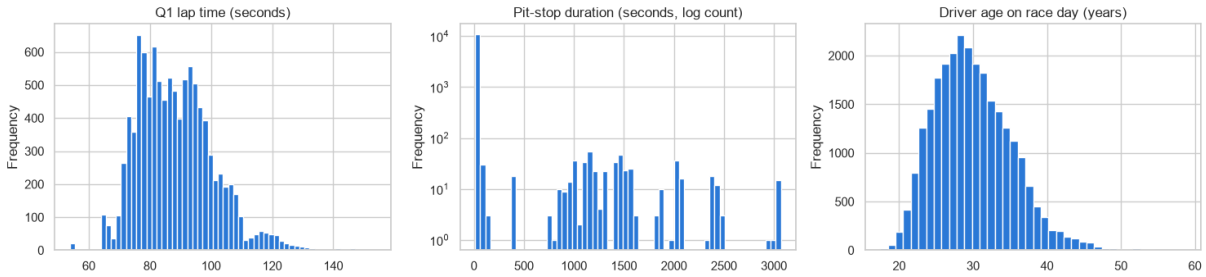


Figure 4. *Distributions that only exist after typing: Q1 lap time, pit-stop duration (log count) and driver age.*

The cleaning did not distort the target. It removed 91 rows and 2 podium rows, and both of those were drivers counted twice in a shared car, so no driver lost a podium. The podium rate moves from 12.69% to 12.73%, and the per-season rate is identical from 1979 onward.

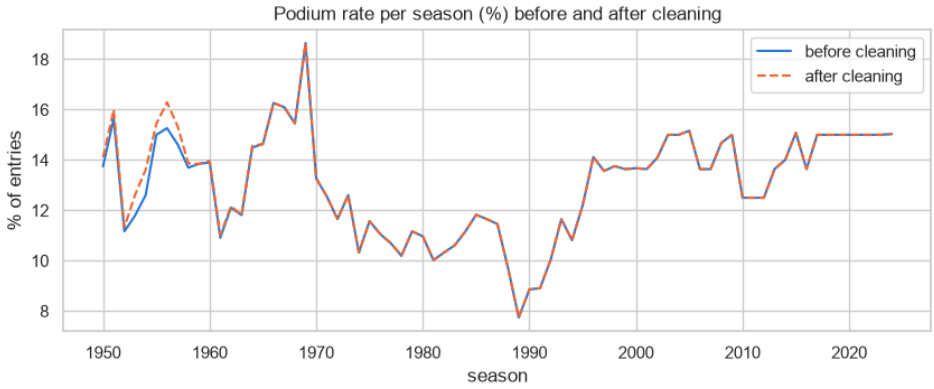


Figure 5. *Podium rate per season before and after cleaning.*

1.4 Modelling window and split

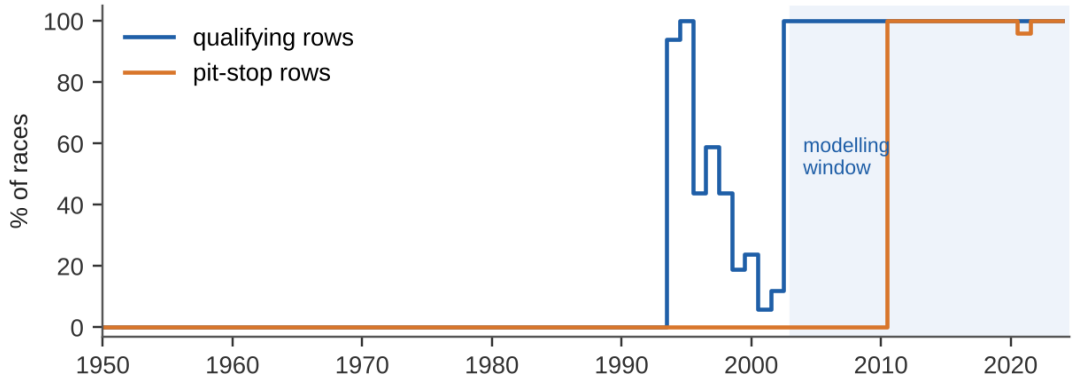


Figure 6. *Share of races that have qualifying and pit-stop rows. Qualifying is complete only from 2003; 1996–2002 are patchy (as low as 6% in 2001). This sets the modelling window.*

The modelling table covers 2003–2024, the seasons with complete qualifying. It has 8,906 rows (drivers who actually started) in 428 races, each race with exactly three podium rows, so the positive rate is 14.4%. Older seasons are not thrown away: they still feed the history behind the form features. We split by season rather than at random, so a race never ends up in two sets and the test seasons are genuinely in the future: training 2003–2018 (6,362 rows), validation 2019–2021 (1,193) and test 2022–2024 (1,351). A random row split would have put drivers from every test race into training [3.3].

2. Data-engineering questions

No single table answers these questions, so we built one joined table, `entries`, and reused it for all three: `results` joined to `races` (year, circuit), `circuits` (name, country), `drivers` (nationality), `qualifying` (qualifying position, left join), `status` (retirement reason) and `constructors` (team name). An assert confirms the joins neither add nor drop rows, so the grain stays one row per driver per race.

2.1 Which circuits turn a front-row start into a podium most often, and did that change after 2014?

We counted front-row starts in two ways because they are not the same thing: by grid (`grid` 1 or 2) and by qualifying (`quali_position` 1 or 2). Penalties can push a front-row qualifier back, and others move up to fill the slot. The conversion rate is podiums divided by front-row starts per circuit and era, comparing 2003–2013 with the hybrid era 2014–2024. Both eras have full qualifying data and are about 11 seasons long. A circuit is ranked in an era only with at least 10 front-row starts, since below that one race moves the rate by 10 points or more.

Answer. Front-row starts convert more often in the hybrid era: 68% before 2014 and 76% after by grid (67% to 74% by qualifying). In 2014–2024 the best circuits are Suzuka (89%, 16 of 18), Yas Marina (86%) and Monaco (85%), followed by Spa and Barcelona (82%); the Hungaroring is last at 59%. Before 2014 the leaders were Magny-Cours (92%) and Istanbul Park (79%), then Monaco and Monza (77%). Those first two do not have enough front-row starts after 2014 to be ranked.

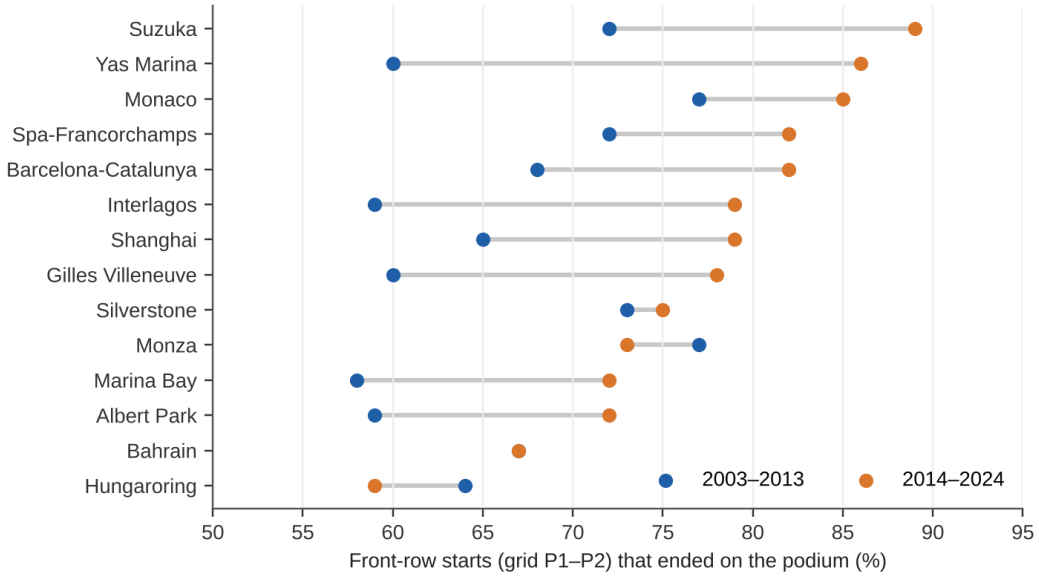


Figure 7. *Conversion rate by circuit, for the 14 circuits with at least 10 front-row starts in both eras (grid definition). Sorted by the 2014–2024 rate.*

The ranking changed a lot. Across the 14 circuits ranked in both eras the Spearman correlation between the two rankings is only 0.30 ($p = 0.29$). Eleven of the 14 circuits convert better after 2014. Bahrain stayed at 67% but still lost six places, because nearly everyone else improved. Table 1 shows every circuit's move.

Table 1. *Rank changes between eras, grid definition, sorted by size of the move (ties share a rank).*

Circuit	Rate 2003–13	Rank	Rate 2014–24	Rank	Places moved
Monza	77%	1	73%	10	−9
Yas Marina	60%	10	86%	2	+8
Bahrain	67%	7	67%	13	−6
Interlagos	59%	12	79%	6	+6
Silverstone	73%	3	75%	9	−6
Hungaroring	64%	9	59%	14	−5
Suzuka	72%	4	89%	1	+3
Marina Bay	58%	14	72%	11	+3
Gilles Villeneuve	60%	10	78%	8	+2
Barcelona-Catalunya	68%	6	82%	4	+2
Monaco	77%	1	85%	3	−2
Albert Park	59%	12	72%	11	+1
Shanghai	65%	8	79%	7	+1
Spa-Francorchamps	72%	4	82%	4	0

Grid versus qualifying makes a small difference to the rates and almost none to the order. 74 entries differ between the definitions in 2003–2024 (36 started on the front row without qualifying there, 38 qualified there and started further back). The rate changes at 6 of 21 circuits before 2014 and 7 of 19 after, but the two rankings correlate at 0.94 and 0.88. Table 2 shows where the top five differ. The top circuit in each era is the same under both definitions (Magny-Cours, then Suzuka). Below that the order shuffles: before 2014 Monaco drops out of the qualifying top five and Spa comes in, and after 2014 Yas Marina falls from 2nd to 5th. The qualifying definition gives slightly lower rates in both eras, which fits front-row qualifiers who were moved back converting less often.

Table 2. *Top five circuits per era under each front-row definition.*

	2003–13, grid	2003–13, qualifying	2014–24, grid	2014–24, qualifying
1	Magny-Cours 92%	Magny-Cours 92%	Suzuka 89%	Suzuka 89%
2	Istanbul Park 79%	Istanbul Park 79%	Yas Marina 86%	Monaco 85%
3	Monza 77%	Silverstone 77%	Monaco 85%	Barcelona 82%
4	Monaco 77%	Monza 73%	Barcelona 82%	Spa 82%
5	Silverstone 73%	Spa 72%	Spa 82%	Yas Marina 82%

Grain, missingness, cutoff. Each rate aggregates driver-race rows, two front-row rows per race. `grid` is never missing and `grid = 0` is never treated as a front-row start. Qualifying position is missing for 24 of 8,970 entries, which only drop out of the qualifying definition. Grid and qualifying are both known before the start; the podium is the outcome, which is fine for a descriptive question. Each rate rests on 10–24 starts, so gaps of a few points are noise.

2.2 Do drivers podium more often at home, once starting position is controlled for?

A race is a home race when the driver's nationality matches the *circuit's* country, not the race name (Imola's San Marino GP is a home race for Italians). Dual nationalities count as home in both countries, and every nationality in the data is either mapped or listed as having no home circuit (checked by an assert). We use drivers who started from a real grid slot (`grid > 0`), over all seasons, without the eleven Indianapolis 500 races of 1950–1960, whose fields were almost entirely Americans who never raced another Grand Prix. That leaves 24,499 starts, of which 1,923 (7.8%) were at home.

The naive comparison points the wrong way: 12.0% podiums at home versus 13.9% away. Home drivers simply start further back more often (59% of home starts are from P11 or lower, against 54% of away starts).

So we compare within grid groups, then summarise with a logistic regression of podium on home plus grid group, with standard errors clustered by race because drivers in the same race compete for the same three places.

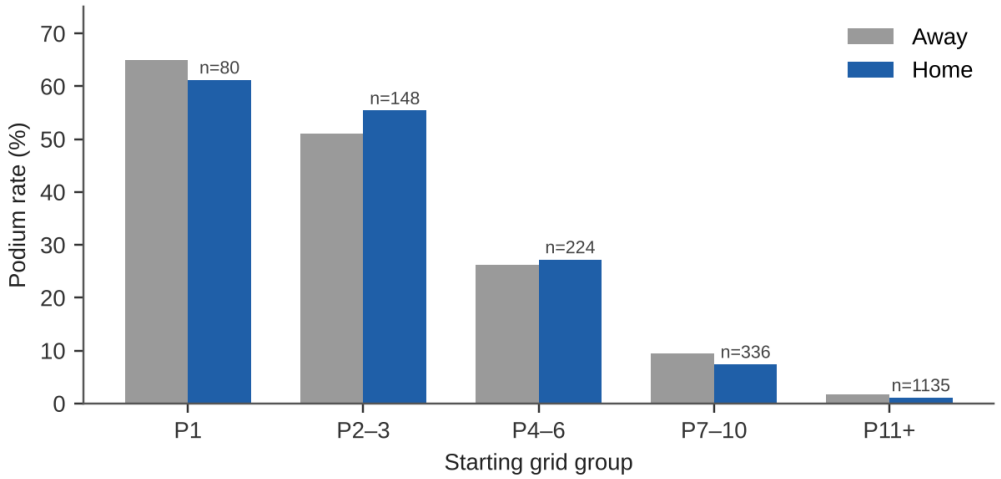


Figure 8. *Podium rate at home and away within each grid group, all seasons. n is the number of home starts in the group.*

Table 3. *Logistic regression podium ~ home + grid group, race-clustered standard errors. An odds ratio of 1 means no difference between home and away.*

Sample	Starts	Home starts	Odds ratio (home)	95% CI	p-value
All seasons (main answer)	24,499	1,923	0.954	0.808 – 1.126	0.577
2003–2024 (check)	8,837	363	0.778	0.523 – 1.158	0.217

Answer: no clear home advantage. Within grid groups the difference flips sign: home drivers do a little better from P2–3 (+4.3 points) and P4–6 (+1.1), and a little worse from P1 (−3.7), P7–10 (−1.9) and P11+ (−0.5). Over all seasons the home odds ratio is 0.95, and the interval runs from about 19% lower to 13% higher odds. The 2003–2024 check agrees but is wider, since it rests on only 363 home starts. Both intervals contain 1. This is absence of evidence rather than proof of no effect, and we only controlled for grid position, not for team or era.

Grain, missingness, cutoff. One row is one race start. Grid, nationality and circuit country are never missing; 2,169 rows are excluded on purpose (non-starters, `grid = 0`, Indianapolis 500). Nationality and circuit country are known well before the race, so a home flag would be a legal feature; it was tested as `home_race` and dropped (Section 3).

2.3 Which constructors had the highest mechanical-retirement rate in 2014–2021 and 2022–2024?

The retirement reason lives in `status`. We put each of the 66 status codes used in 2014–2024 into one category: finished, mechanical (41 codes, e.g. Engine, Power Unit, Gearbox, Hydraulics, plus failed car parts such as Front wing), crash or contact, driver (Illness), regulation (Disqualified, Excluded), did not start, pit-stop or team error (Wheel nut, Out of fuel), tyre, and unknown (plain "Retired"). Only the mechanical category counts. The rate is mechanical retirements divided by race starts per team and period; non-starters are not starts, pit-lane starters are. A team is ranked only with at least 40 starts in a period.

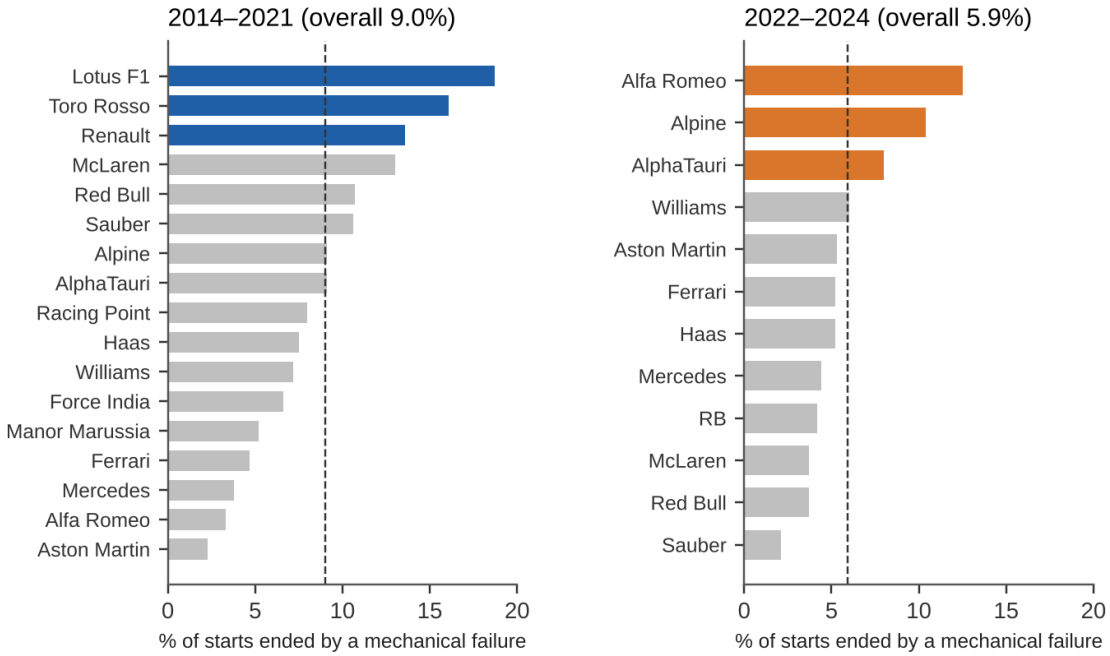


Figure 9. *Mechanical-retirement rate per constructor and period (teams with at least 40 starts). The three highest in each period are coloured; the dashed line is the overall rate.*

Table 4. *Highest mechanical-retirement rates (mechanical retirements / starts).*

Rank	2014–2021	2022–2024
1	Lotus F1 18.7% (14 / 75)	Alfa Romeo 12.5% (11 / 88)
2	Toro Rosso 16.1% (39 / 242)	Alpine 10.4% (14 / 135)
3	Renault 13.6% (27 / 199)	AlphaTauri 8.0% (7 / 88)
4	McLaren 13.0% (41 / 315)	Williams 6.0% (8 / 134)
5	Red Bull 10.7% (34 / 318)	Aston Martin 5.3% (7 / 133)

Answer. Lotus F1, Toro Rosso and Renault had the highest rates in 2014–2021; Alfa Romeo, Alpine and AlphaTauri in 2022–2024. Mechanical failures became rarer overall, from 9.0% of starts to 5.9%. Among the 11 teams present in both periods, McLaren (13.0% to 3.7%), Sauber (10.6% to 2.1%) and Red Bull (10.7% to 3.7%) improved most, while Alfa Romeo got much worse (3.3% to 12.5%). In 2022–2024 the four front-running teams are all at or below 5.2%, though the lowest rate belongs to Sauber, so reliability is not only a top-team trait.

"Retired" has no cause attached, so we left it out of the mechanical count. To check that the answer does not depend on that choice, we recounted with the 50 "Retired" codes treated as mechanical, which is the upper bound (Table 5). The top team in each period stays the same and Toro Rosso stays second before 2022. McLaren replaces Renault in third before 2022. After 2022, RB rises to joint second, but only because it had 3 "Retired" codes in just 48 starts. So the leaders are robust and the order lower down is sensitive for teams with few starts.

Table 5. *Robustness check: top three with and without "Retired" counted as mechanical.*

	2014–21, main	2014–21, "Retired" counted	2022–24, main	2022–24, "Retired" counted
1	Lotus F1 18.7%	Lotus F1 22.7%	Alfa Romeo 12.5%	Alfa Romeo 12.5%
2	Toro Rosso 16.1%	Toro Rosso 17.8%	Alpine 10.4%	RB 10.4%
3	Renault 13.6%	McLaren 14.6%	AlphaTauri 8.0%	Alpine 10.4%

Grain, missingness, cutoff. Each row is one car's start, so a two-car team has two rows per race. `status` is never missing and every code used has a category (assert). The current race's status is only known after the

race, so it can never be a feature; a team's mechanical-failure rate over its *previous* races is legal, and became `team_mech_dnf_rate`. Post-2022 counts are small (one failure moves a rate by 1–2 points), and the same team can appear under several names (Toro Rosso, AlphaTauri, RB) because we kept the dataset's constructor ids.

3. Selected features and their validity at the cutoff

The prediction is made after qualifying and before the start, so a field is usable only if it exists at that moment. Two fields are easy to get wrong. `results.grid` sits in the results table but is the official starting grid, so it is legal. `driver_standings` for the same `raceId` looks like "standings before this race", but it is the table *after* the race and already includes its result. A quick single-feature check shows the difference: the same-race championship position scores ROC-AUC 0.93 against the podium, the previous round's 0.89, and current-race points 0.97. The rule we followed is that current-race values come only from `qualifying` and `grid`; every other result field enters through a `shift(1)` over earlier races or through the previous round's standings [3.1].

3.1 Candidates and design trials

We built 17 candidates in six families (qualifying, grid, standings, form, reliability, context). Two design choices were settled by trials on the training seasons. The first was the form window. We tried 1, 3, 5, 10 and 20 races: the score rises steeply up to 5 races and gains under 0.01 ROC-AUC from 10 to 20, so we use 10 races. That is about half a season, and it still reacts reasonably fast when a driver changes team. The second was the qualifying gap. Measuring it from each driver's best lap in any session beat Q1 only (ROC-AUC 0.89 against 0.83, PR-AUC 0.59 against 0.45), because front-runners often do just enough in Q1 to get through.

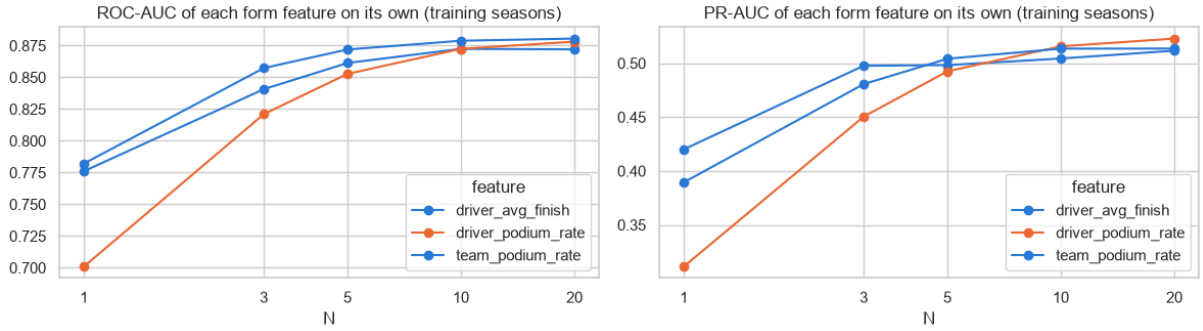


Figure 10. *Form-window trial: ROC-AUC and PR-AUC of each form feature on its own, for windows of 1–20 races (training seasons).*

3.2 Selection rule and evidence

We selected on the training seasons only. A candidate was dropped if its ROC-AUC on its own was below 0.55, or if it had |Spearman ρ| ≥ 0.9 with a stronger feature already kept. Spearman is used because the relationships are monotonic but clearly not linear.

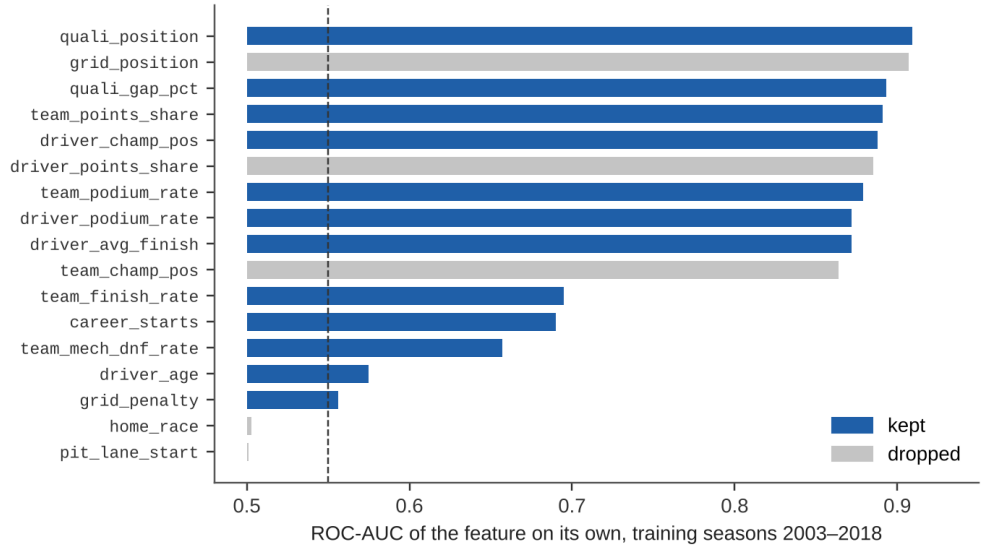


Figure 11. *Each candidate scored on its own on the training seasons. Dashed line: the 0.55 relevance threshold. Grey features were dropped.*

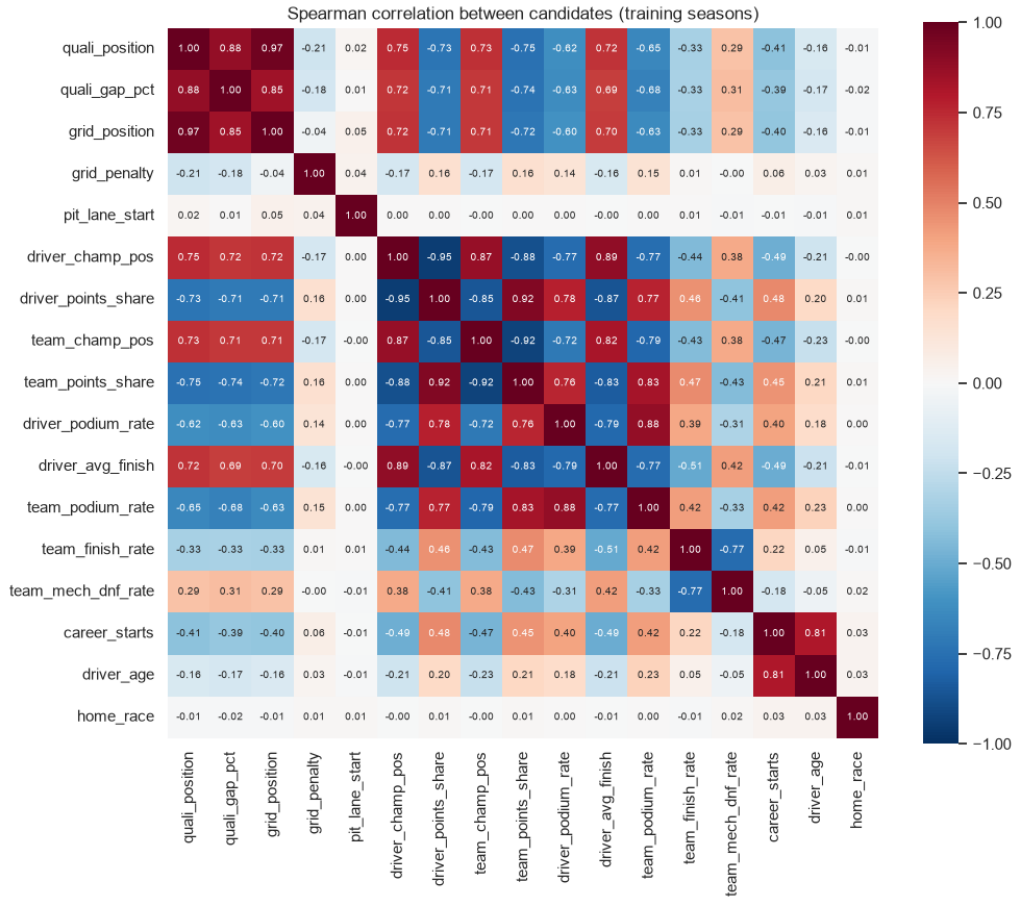


Figure 12. *Spearman correlation between all 17 candidates, training seasons. Four pairs reach $|\rho| \geq 0.9$.*

The heatmap shows four pairs that are close to copies of each other. Grid and qualifying position are at 0.97, because the grid simply is the qualifying order for most drivers. Driver championship position and driver points share are at −0.95, team championship position and team points share at −0.92, and driver and team points share at 0.92, since teammates share a car. Form and standings also correlate (0.78–0.89), but they are not copies: standings cover the current season and form covers the last 10 races, so both stay.

3.3 The selected set

Table 6. *The 12 selected features.*

Group	Feature	Definition	Why it is known before the start
Qualifying	`quali_position`	Qualifying position	Qualifying is over
	`quali_gap_pct`	Best lap in Q1–Q3 vs. fastest lap of the session, in %	Qualifying is over
Grid	`grid_penalty`	Grid slot minus qualifying position (pit lane = behind last slot)	Official grid and qualifying are both fixed before the start
Standings	`driver_champ_pos`	Driver's championship position	Taken after the previous round
	`team_points_share`	Team points ÷ leader's points	Taken after the previous round
Form	`driver_podium_rate`	Podium share, driver's last 10 races	Strictly earlier races (shift(1))
	`driver_avg_finish`	Mean finishing position, last 10 races	Strictly earlier races
	`team_podium_rate`	Podium share of team's cars, last 10 races	Strictly earlier races
Reliability	`team_finish_rate`	Share of team's cars that finished, last 10 races	Strictly earlier races
	`team_mech_dnf_rate`	Share with a mechanical retirement, last 10 races	Strictly earlier races
Context	`career_starts`	Number of previous F1 entries	Count of earlier entries
	`driver_age`	Age on race day	Date of birth and race date

Points *share* is used instead of raw points because the scoring system changed in 2003, 2010 and again with sprints in 2021, so raw totals are not comparable across seasons. A driver who joins mid-season is placed behind everyone listed with a share of 0, rather than left missing.

`grid_penalty` is the one feature kept despite a weak score on its own (0.56), since 73% of rows have no penalty. Its value is conditional: among drivers who qualified in the same place, those moved back reach the podium far less often (Figure 13). None of the four pole-sitters who were moved back reached the podium, against 79% of those who kept pole. It also means `grid_position` is not lost: `quali_position` + `grid_penalty` equals the grid slot, and the pair is almost uncorrelated ($\rho = -0.21$) where grid and qualifying are near copies (0.97).

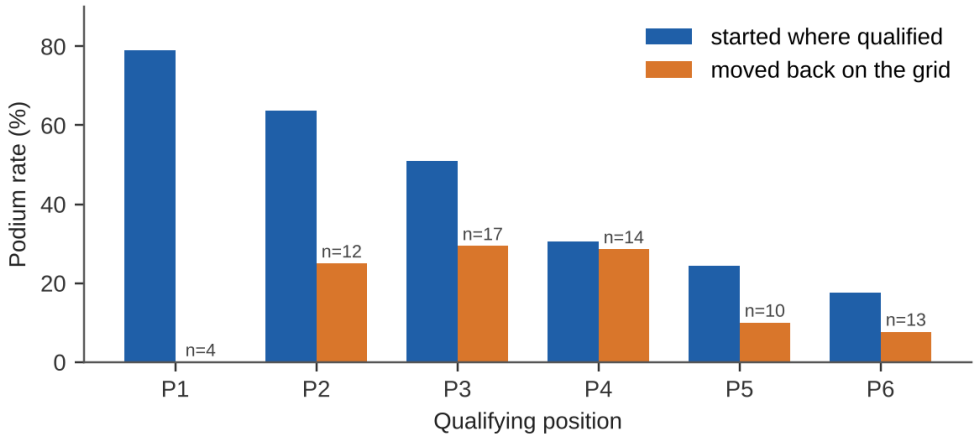


Figure 13. *Podium rate by qualifying position for drivers who started where they qualified versus those moved back (training seasons). n is the size of the moved-back group, so single bars are noisy.*

Dropped: `grid_position` ($\rho = 0.97$ with qualifying position), `driver_points_share` (0.95 with driver championship position, 0.92 with team points share), `team_champ_pos` (0.92 with team points share), `home_race` (no signal, consistent with 2.2) and `pit_lane_start` (no signal, only 8 of 6,362 training rows, and already encoded in `grid_penalty`). Driver, team and circuit ids are deliberately not features: an id says

nothing stable about speed, and the test seasons include drivers and teams the model never saw.

3.4 Missing values and preprocessing

Every missing value in the modelling table has a structural reason, and each family is missing on exactly the same rows. That gives only five distinct reasons (Table 7).

Table 7. *Missing values per split and their reason.*

Feature(s)	Missing because	Train	Val.	Test
Standings (driver_champ_pos, team_points_share)	Round 1: no previous standings	5.2%	5.0%	4.4%
driver_podium_rate, driver_avg_finish	Driver's first F1 race	1.1%	0.8%	0.6%
Team form and reliability (3)	Team's first race under this constructor id	0.5%	0.5%	0.1%
quali_position, grid_penalty	No qualifying row	0.3%	0.3%	0.0%
quali_gap_pct	No qualifying time set	1.7%	1.7%	1.0%

Gaps are filled with the training median, plus one missing-value flag per reason (five flags, giving 17 model inputs). Because each family is missing on the same rows, one flag per column would just duplicate columns. Then everything is standardised. All of these steps are fitted on the training seasons only. Only one feature gets a log transform (Table 8). The qualifying gap has a long tail that reaches 38% in wet or disrupted sessions, and `log1p` pulls it in while keeping pole at 0. `grid_penalty` is just as skewed, but it is signed and its big values are real events, so it is only scaled. The rates are bounded between 0 and 1, so there are no extreme values to tame.

Table 8. *Skewness of the selected features on the training seasons, and how each is treated.*

Feature	Skew	Min	Max	Treatment
grid_penalty	4.91	−7	23	scaled only (signed)
quali_gap_pct	4.63	0	37.8	log1p, then scaled
driver_podium_rate	1.72	0	1	scaled only (bounded)
team_mech_dnf_rate	1.59	0	1	scaled only (bounded)
team_podium_rate	1.57	0	1	scaled only (bounded)
team_points_share	1.00	0	1	scaled only
career_starts	0.83	0	325	scaled only
driver_age	0.49	17.5	43.9	scaled only
driver_champ_pos / quali_position	0.15 / 0.06	1	28 / 24	scaled only
driver_avg_finish / team_finish_rate	−0.11 / −0.93			scaled only

4. Models, tuning and evaluation

All models use the same rows and the same 17 inputs, so their scores are directly comparable. Because only about 14% of rows are podiums, accuracy is not useful: a model that never predicts a podium already scores 85–86%. Our **primary metric is PR-AUC** (average precision). It measures how well the rare podium class is ranked above the rest, and it does not depend on a threshold. The **secondary metric is F1** at a threshold chosen on validation, which measures the quality of the actual yes/no call. ROC-AUC, precision, recall and accuracy are reported as well. Hyperparameters were chosen by validation PR-AUC. Each model's threshold was then chosen to maximise validation F1 and frozen. The test set was scored once, after everything was fixed.

4.1 Tuning

Baseline logistic regression: L2 penalty with C = 1, no class weights. **Enhanced logistic regression:** we searched C over {0.01, 0.1, 1, 10} and class weights over {none, balanced}, eight configurations in total. C = 10 without weights won, but only by 0.0002 PR-AUC over the baseline, so tuning changed practically nothing. Balanced weights were worse at every C (about 0.012 lower). **XGBoost:** we searched max_depth over {2, 3}, min_child_weight over {1, 5, 10} and scale_pos_weight over {1, 6.07}, twelve configurations. The training negative-to-positive ratio is 6.07. Each configuration uses learning rate 0.05, subsample and colsample 0.8, and early stopping on validation with patience 50.

Table 9. *Enhanced logistic regression grid.*

Config	Val. ROC-AUC	Val. PR-AUC	Config	Val. ROC-AUC	Val. PR-AUC
C 0.01	0.935	0.717	C 0.01, balanced	0.935	0.713
C 0.1	0.939	0.733	C 0.1, balanced	0.937	0.723
C 1 (baseline)	0.939	0.736	C 1, balanced	0.937	0.724
C 10 (selected)	0.939	**0.736**	C 10, balanced	0.937	0.725

Table 10. *XGBoost grid. The selected configuration (bold) edges out depth 2 / weight 1 by 0.0004 PR-AUC; trees are the early-stopped count.*

Depth	Min child wt.	Pos. weight	Trees	Val. ROC-AUC	Val. PR-AUC
2	1	1	104	0.941	0.751
2	1	6.07	96	0.940	0.739
2	5	1	102	0.941	0.749
2	5	6.07	135	0.940	0.743
2	10	1	101	0.941	0.746
2	10	6.07	105	0.940	0.746
3	1	1	34	0.938	0.743
3	1	6.07	95	0.940	0.743
3	**5**	**1**	**68**	**0.939**	**0.751**
3	5	6.07	28	0.939	0.737
3	10	1	74	0.940	0.742
3	10	6.07	7	0.935	0.735

In both models the unweighted versions did better on validation PR-AUC, so we kept the real 14% class balance and handled the imbalance through the threshold instead. **Shallow neural network (Keras):** 17 → 32 → 16 → 1 with ReLU hidden layers and a sigmoid output, 1,121 parameters. It uses L2 1e-4 on both hidden layers, Adam with learning rate 0.001, batch size 64, and early stopping on validation loss with patience 20, restoring the best weights. It stopped after 44 epochs, with the best epoch at 24.

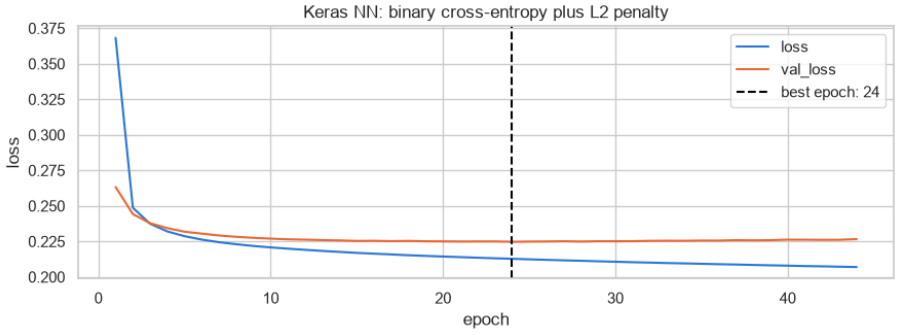


Figure 14. *Keras network: training and validation loss per epoch; the dashed line marks the best epoch.*

4.2 Results

Table 11. *All metrics for every model on train, validation and the unseen test seasons. The threshold was chosen on validation and applied unchanged to every split.*

Model	Split	ROC-AUC	PR-AUC	F1	Prec.	Recall	Acc.	Thresh.
Baseline LR	train	0.930	0.687	0.659	0.547	0.830	0.879	0.231
	validation	0.939	0.736	0.736	0.639	0.867	0.906	
	test	0.933	0.697	0.652	0.517	0.882	0.858	
Enhanced LR	train	0.930	0.686	0.658	0.545	0.830	0.878	0.230
	validation	0.939	0.736	0.734	0.637	0.867	0.905	
	test	0.933	0.697	0.653	0.517	0.887	0.858	
XGBoost	train	0.943	0.723	0.682	0.639	0.732	0.904	0.369
	validation	0.939	**0.751**	0.737	0.688	0.794	0.915	
	test	0.935	0.691	0.661	0.621	0.706	0.891	
Keras NN	train	0.941	0.713	0.678	0.607	0.768	0.897	0.322
	validation	0.933	0.708	0.726	0.668	0.794	0.910	
	test	0.930	0.702	0.636	0.557	0.740	0.872	

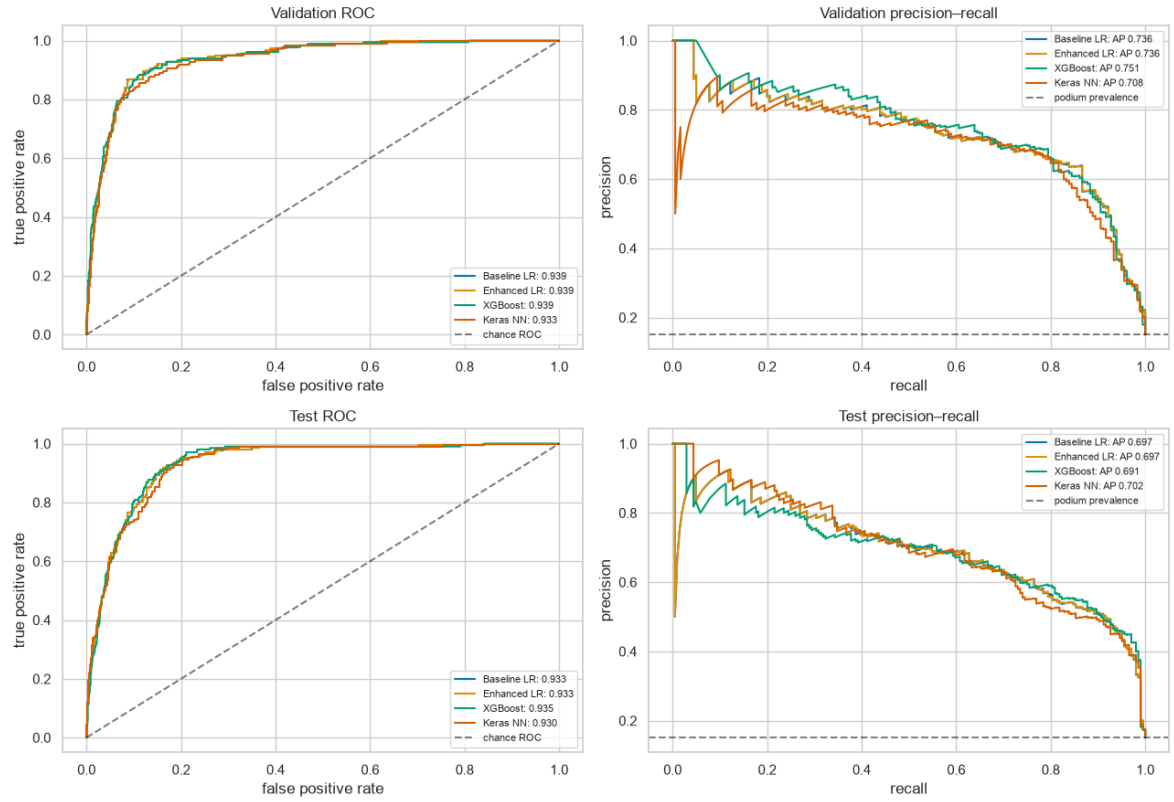


Figure 15. *ROC and precision–recall curves on validation (top) and test (bottom).*

XGBoost had the best validation PR-AUC (0.751 against 0.736 for the baseline), so the protocol selected it before the test set was touched. On test the models bunch together: all four are within 0.012 PR-AUC, and the network is actually highest at 0.702. XGBoost keeps the best test F1 (0.661) and ROC-AUC (0.935). These margins are smaller than the drop from validation to test (0.04–0.06 PR-AUC), so our reading is that the pre-race inputs carry most of the available signal and a linear model already captures nearly all of it. XGBoost stays the chosen model because swapping it out after seeing test scores would be selecting on the test set.

5. Experiments

5.1 Feature-group ablation

We retrained XGBoost with the selected settings six times, each time removing one feature group, and compared against the full model on validation. Every group costs some PR-AUC. Qualifying costs by far the most, with a drop of 0.089 PR-AUC and 0.040 F1, followed by grid at 0.035. Form, standings, context and reliability each cost 0.011–0.019. The small drops for form and standings fit the overlap seen in the heatmap: when one group is removed, the other still carries much of the same information. These drops measure what the model relies on, given the other groups. They are not causal effects.

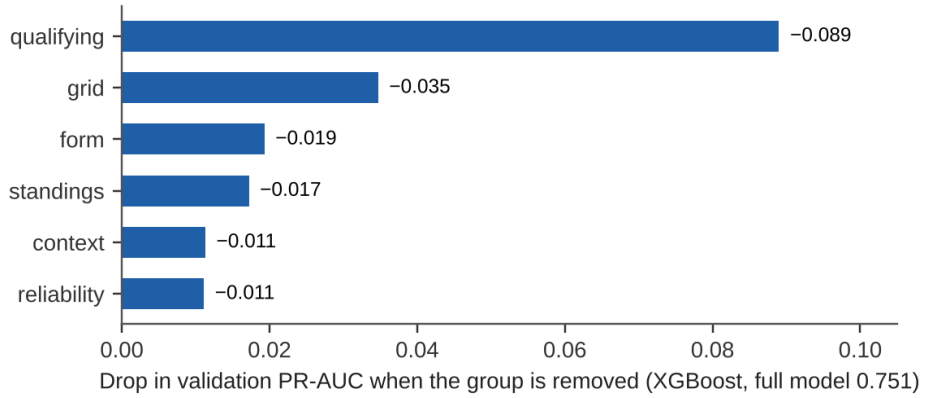


Figure 16. *Feature-group ablation on the validation seasons (XGBoost with fixed settings).*

5.2 Preprocessing-order experiments

We also changed the order of the numeric steps in the enhanced logistic regression, keeping everything else fixed. Order 1 imputes before the log transform. The inputs came out identical (max difference 0), because log1p preserves order and so the median lands on the same row. Order 2 scales before imputing. This changes the inputs by up to 0.07 standard deviations, because the scaler's statistics are then computed from observed values only. The validation score barely moved (−0.00007 PR-AUC). So the order does not matter for this data. We kept the reference order, which fills gaps before computing the scaling statistics.

Table 12. *Preprocessing-order experiments, enhanced logistic regression, validation seasons.*

Experiment	Order	Max input change	Val. PR-AUC	Change	Val. F1
Reference	log → impute → scale	0	0.73642	—	0.7341
Order 1	impute → log → scale	0	0.73642	+0.00000	0.7341
Order 2	log → scale → impute	0.069	0.73635	−0.00007	0.7341

6. Explainability

We explain the selected XGBoost model on the validation seasons, without retraining it. **None of these explanations are causal.** A feature being important means the model relies on it to predict, not that it causes podium finishes.

6.1 Global: permutation importance and SHAP

Permutation importance shuffles one input at a time and measures the drop in validation PR-AUC, averaged over 20 repeats. Qualifying position dominates, with a drop of 0.204. Next come the qualifying gap (0.035), driver podium rate (0.019) and `grid_penalty` (0.019). The missing flags and `team_mech_dnf_rate` are close to zero. That does not make them useless: on validation, team reliability overlaps with team strength, so shuffling it alone loses little.

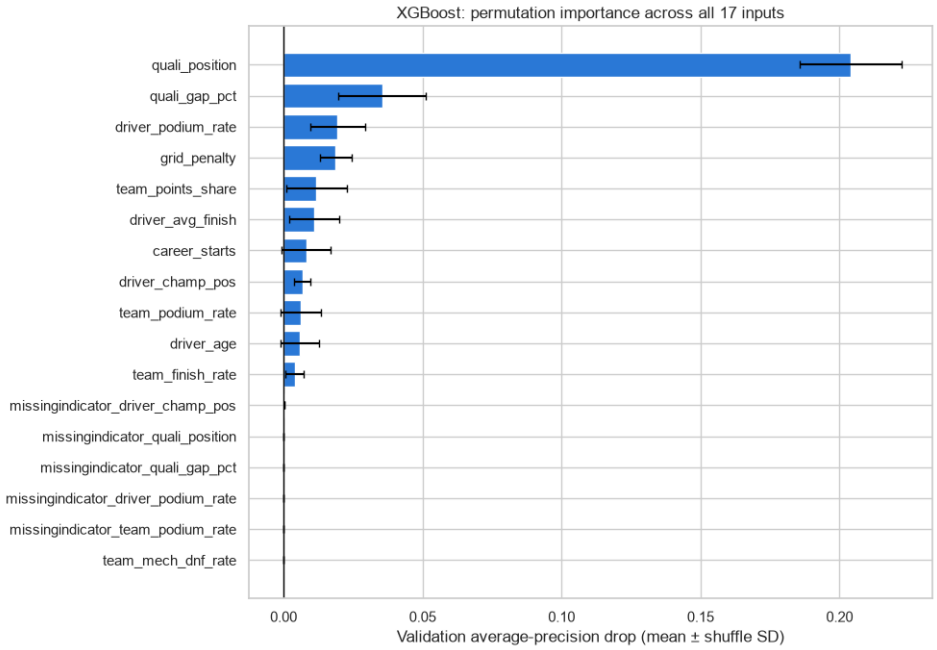


Figure 17. *Permutation importance on validation: mean drop in PR-AUC over 20 shuffles (error bars show shuffle variability, not confidence intervals).*

SHAP measures how far each input moves a prediction, in log-odds, rather than how much performance is lost. The order is broadly the same. Qualifying position leads (mean |SHAP| 1.03), then the qualifying gap (0.35), team points share (0.32) and driver podium rate (0.28). The beeswarm shows the expected directions. A good qualifying position, a small gap to pole, a high team points share and a high recent podium rate push towards a podium. A positive `grid_penalty` (being moved back) pushes strongly away, even though it moves few predictions on average. The two methods rank the team features differently because they measure different things, and correlated features can share or mask each other's credit.

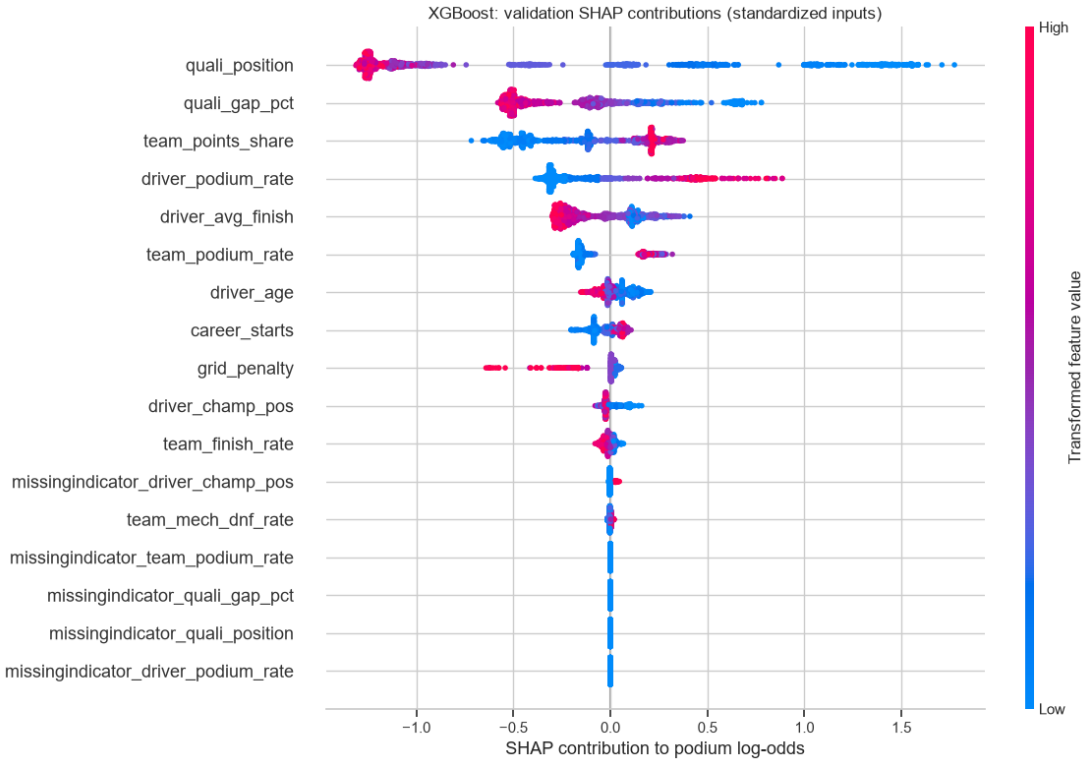


Figure 18. *SHAP beeswarm on validation. Each dot is one driver-race; colour is the (standardised) input value.*

6.2 Local: one prediction, SHAP and LIME

For a local example we took the validation prediction closest to the decision threshold, picked by a fixed rule that does not look at outcomes. It is Sebastian Vettel at the 2019 Monaco GP, with a predicted probability of 36.9% against a threshold of 36.9%. He qualified 4th, had a 40% podium rate over his last 10 races, and his team held 56% of the leader's points. He did finish on the podium. Starting from the model's average of −1.83 log-odds, qualifying 4th adds +0.58, his podium rate +0.29, the team's podium rate +0.20 and the team points share +0.20. A 1.1% gap to pole takes away 0.13.

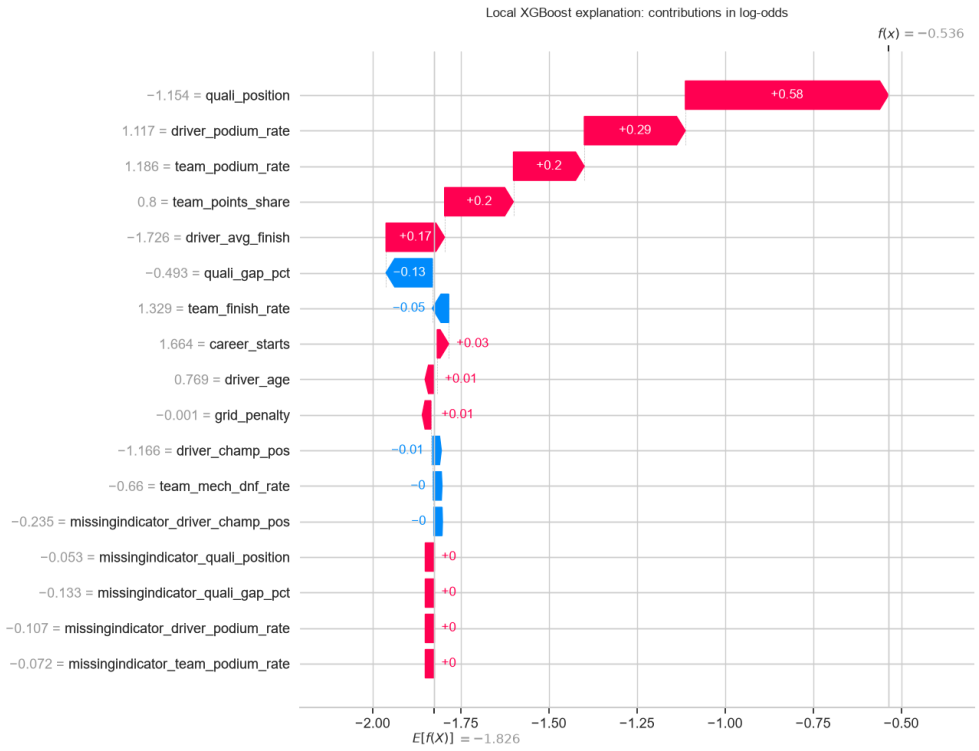


Figure 19. *Local SHAP waterfall for the borderline example (values on the left are standardised inputs).*

LIME explains the same row in a different way. It perturbs the row, scores the copies with the model, and fits a small linear model to them. The surrogate reproduces the model's 36.91% exactly, with a local fit R^2 of 0.68. All five of the strongest SHAP contributions point the same way in LIME (Table 13), which gives us some confidence in the local explanation. It still describes the model's reasoning, not a cause.

Table 13. *Leading local contributions under both methods.*

Feature	Local SHAP (log-odds)	LIME weight (probability)	Same direction
quali_position	+0.576	+0.187	yes
driver_podium_rate	+0.287	+0.057	yes
team_podium_rate	+0.201	+0.025	yes
team_points_share	+0.195	+0.055	yes
driver_avg_finish	+0.166	+0.034	yes

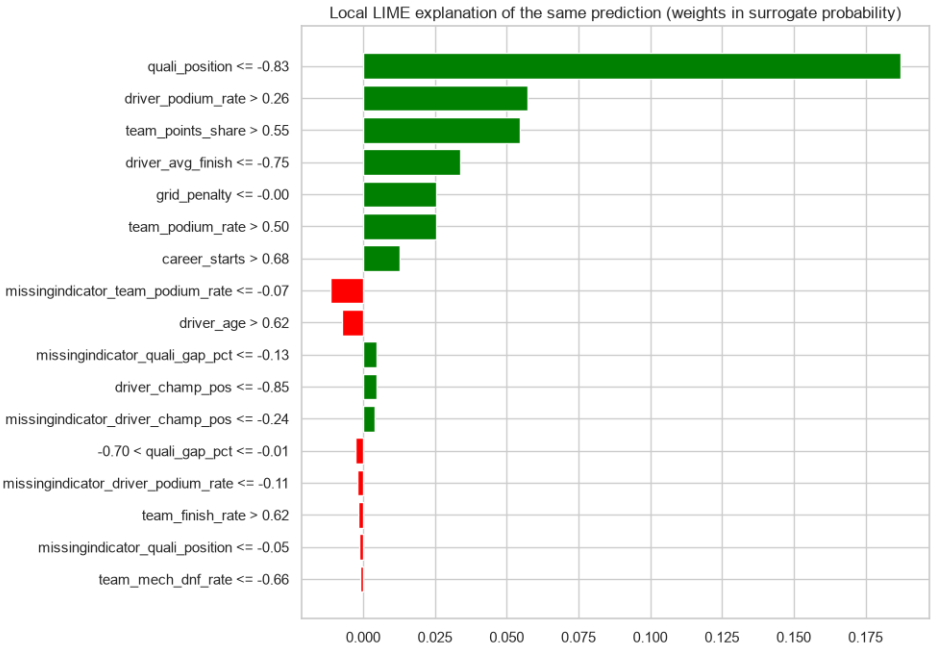


Figure 20. *LIME weights for the same prediction.*

7. Inference function

`predict_podium(record)` takes one driver-race as a dictionary or Series of the 12 engineered, unscaled features. It applies the training-fitted pipeline (medians, the five missing flags, log1p and scaling) and the frozen XGBoost model and threshold. It returns the probability, the yes/no prediction and a list of the missing features. Omitted keys, None, NaN and `\N` all count as missing. Unknown keys such as an accidental outcome field, infinite values and empty records are rejected. We demonstrated it on Lewis Hamilton at a 2019 race (raceId 1011), first with the complete record, then with the qualifying gap and both standings features removed.

Table 14. *Inference on a complete and a partially missing record.*

	Complete record	Partially missing
quali_position / grid_penalty	3 / 0	3 / 0
quali_gap_pct	0.37%	missing
driver_champ_pos / team_points_share	2 / 1.00	missing / missing
Form (driver podium rate, avg. finish, team podium rate)	0.90, 1.7, 0.65	same
Reliability, career starts, age	1.00, 0.00, 230, 34.2	same
Podium probability	**0.606**	**0.392**
Prediction (threshold 0.369)	podium	podium

Both calls predict a podium. Without the qualifying and standings information, though, the probability falls by 0.21, landing just above the threshold. The function works with gaps, but it is far less confident without qualifying data. Its checks confirm that the complete call reproduces the stored validation probability exactly.

8. Known limitations

Qualifying coverage drift. Qualifying rows exist from 1994 but are complete only from 2003 (Figure 6), and the qualifying format itself changed: a single lap in 2003–2005, three knock-out sessions after. This forced the 2003 start and means the model cannot be applied to older seasons without imputing its most important inputs. We checked whether a longer history would help anyway: training from 1950 to 2018 lowered logistic regression's validation PR-AUC by about 0.03 and left XGBoost within noise (0.739 against 0.751), because the old rows add form and standings from a different sport, not more qualifying signal [4.8]. The same gap matters at inference time, as Section 7 showed (0.61 to 0.39).

Era sensitivity. The test seasons begin with the 2022 ground-effect regulations. All models lose PR-AUC from validation to test (XGBoost 0.751 to 0.691, logistic regression 0.736 to 0.697; Table 11). Reliability also drifted sharply: the team finish rate is around 0.56 in older training rows and 0.85 in 2019–2024, and mechanical retirements fell from 9.0% to 5.9% between the two periods in 2.3. A model trained mostly on 2003–2018 reads reliability and form features with an older sense of what is normal. In practice it would need retraining as new seasons arrive.

Small samples in the descriptive answers. Circuit conversion rates rest on 10–24 front-row starts and the post-2022 team rates on 48–136 starts, so one race or one failure can move a team or circuit several places. The rank correlation of 0.30 in 2.1 is not distinguishable from zero. The home-race test after 2003 has only 363 home starts.

Definitions that involve judgement. The mechanical category is our own mapping of status codes; "Retired" has no cause and was excluded (tested both ways). "Home" uses nationality, so a driver living abroad still counts as home in his passport country, and countries with several races give their drivers more home starts. Team names follow the dataset's constructor ids, so renamed teams split into separate rows and lose their form history at the rename.

Data quirks. For the 85 shared-drive pairs, the kept row's grid and team belong to the best-result car, which is not always the car the driver started in. This only affects rows before 1979 and so not the model. Red-flag pit stops of 400–3,069 s are real records and were kept; pit stops are not used as features.

Statistical caveats. Rows within a race are dependent, since three podium places are shared. We handled that with race-clustered errors in 2.2 but not in model scoring. Validation is one block of three seasons rather than a rolling cross-validation, it was reused for tuning, early stopping and the threshold, the network was trained with a single seed, and predicted probabilities were not calibrated. Binary predictions also do not force exactly three podiums per race. Feature importance, ablation drops, SHAP and LIME values and the associations in Section 2 describe what the data and the model rely on. None of them shows that a feature causes podium finishes.